

# classify\_to\_functional\_type.R

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| classify_to_functional_type                        |
| <i>Classify Community Data to Functional Types</i> |

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## Description

Classifies taxa in a long-format community data frame to functional-type (FT) labels using a pre-computed FT classification table. Aggregates pollen proportions by all non-‘taxon’/‘value’ identifier columns plus functional type. Produces the same output column structure as ‘classify\_taxonomic\_resolution()’ so it can be used as a drop-in replacement downstream.

## Usage

```
classify_to_functional_type(  
  data_source,  
  data_functional_type_classification,  
  verbose = TRUE  
)
```

## Arguments

|                                     |                                                                                                                                                                                                                                                        |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data_source                         | A data frame containing community data with columns ‘taxon’, ‘dataset_name’, ‘age’, and ‘value’. Other identifier columns such as ‘sample_name’ are preserved.                                                                                         |
| data_functional_type_classification | A data frame mapping taxa to functional types. Must contain columns ‘taxon_name’ (character) and ‘functional_type’ (integer). Typically produced by ‘assign_functional_type_clusters()’ and loaded via ‘load_latest_functional_type_classification()’. |
| verbose                             | Logical. Reserved for informational progress messages. The warning emitted when unmatched taxa are dropped is always shown because it reports data loss.                                                                                               |

**Details**

Steps performed:

1. Validate arguments.
2. Left-join 'data\_source' to 'data\_functional\_type\_classification' on their taxon names.
3. Drop unmatched taxa (NA functional type) with a warning.
4. Create 'taxon' labels "FT\_functional\_type".
5. Aggregate 'value' by original identifier columns and 'taxon'.
6. Full-join back to an identifier/taxon cross-reference to preserve true negatives.

**Value**

A data frame with the same column names as 'data\_source'. The 'taxon' column is replaced by functional-type labels of the form "FT\_1", "FT\_2", etc. 'value' is aggregated (summed) by all original identifier columns and 'taxon'. All identifier combinations present after FT classification are preserved (true negatives kept via a cross-reference join). Taxa not found in 'data\_functional\_type\_classification' are dropped with a 'cli::cli\_warn()' message.

**See Also**

[classify\_taxonomic\_resolution()], [load\_latest\_functional\_type\_classification()], [assign\_functional\_type\_clusters()]

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